Literature Review: Data-Driven Academic Performance Analysis System

Overview

This literature review compiles relevant research papers and resources across eight key areas relevant to the Data-Driven Academic Performance Analysis System project. The papers cover educational data mining, ML prediction models, document parsing with LLMs, topic extraction, learning analytics dashboards, human-in-the-loop systems, and explainability techniques.

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1. EDUCATIONAL DATA MINING & STUDENT PERFORMANCE PREDICTION

Foundation Papers

- **Predicting Students' Academic Performance Using Machine Learning Algorithms** (2021)
  - Comprehensive survey of EDM techniques including support vector regression, Random Forest, and deep learning approaches
  - URL: ScienceDirect (2021)
  - Key insight: Data-driven SPP provides objective reference for education system improvement
- **Educational Data Mining Techniques for Student Performance Prediction: Method Review and Comparison Analysis**
  - Systematic review comparing approaches for predicting student performance
  - Focus on handling diverse factors affecting performance and addressing imbalanced data challenges
  - Covers use cases for students, instructors, and educational managers
  - URL: NCBI/PMC
- **Role of Convolutional Features and Machine Learning for Predicting Student Academic Performance from MOODLE Data**
  - Proposes deep learning system with convoluted features for predicting performance
  - Addresses accuracy, imbalanced data, and feature engineering challenges
  - URL: NCBI/PMC

Advanced Predictive Models

- **Predicting Students' Academic Performance with Conditional Generative Adversarial Network and Deep SVM**

    - Combines CGAN with deep-layer SVM for performance prediction
    - Leverages technology-assisted learning platform data for timely interventions
    - URL: NCBI/PMC
- **The Role of Demographic and Academic Features in Student Performance Prediction**
    - Examines feature selection and model comparison (Decision Trees, SVM, Support Vector Machines achieve 92% accuracy)
    - Discusses how varying features depend on level of study, program, and subject type
    - URL: NCBI/PMC
- **Predicting Students Performance in Educational Data Mining** (IEEE)
    - Applies deep learning classification models for predicting performance
    - Discusses how traditional methods struggle with nonlinear relationships in educational data
    - Addresses the need for models that handle complex dependencies
    - URL: IEEE Xplore

Comprehensive Surveys

- **Predicting Students' Academic Performance to Predict Student Performance Prediction: A Survey Study** (2022)
    - Compares ~260 studies over 20 years
    - Identifies ANN and Random Forest as most-used data mining techniques
    - Discusses major factors affecting performance and feature selection algorithms
    - URL: Springer Nature
- **Educational Data Mining to Predict Students' Academic Performance: A Survey Study** (2022)
    - Focus on identifying at-risk students proactively
    - Comprehensive analysis of data mining tools and techniques
    - URL: Springer Nature (July 2022)
- **Predictive Modelling with OULAD: A Systematic Literature Review** (2024)
    - Reviews predictive modeling using Open University Learning Analytics Dataset
    - Discusses XAI approaches and interpretability in educational context
    - URL: ResearchGate (July 2024)

Specialized Applications

- **Regularized Multi-Path XSENet Ensembler for Enhanced Student Performance Prediction** (2024)
    - Recent ML/data mining advances in education

- Emphasis on sustainability goals and pre-admission data predictions
- URL: NCBI/PMC

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2. DOCUMENT PARSING & EXTRACTION WITH LARGE LANGUAGE MODELS

LLM-Based Document Processing

- **Document Parsing Using Large Language Models - With Code** (Towards Data Science, 2025)
  - End-to-end PDF parsing workflow using GPT-4o and Claude
  - Three-component workflow: Input → Processing → Output
  - Practical implementation using pdfminer.six, PyPDF2, and LLMs
  - URL: Towards Data Science (January 2025)
- **OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations** (2024)
  - Comprehensive benchmark for document content extraction
  - Critical for LLM/RAG systems requiring high-quality document input
  - URL: arXiv (2412.07626)
- **Conversational Text Extraction with Large Language Models Using Retrieval-Augmented Systems**
  - Leverages LLMs for text extraction and user interaction with PDFs
  - Implements Retrieval-Augmented Generation (RAG) for relevant passage retrieval
  - Uses sentence embeddings for document-specific vector stores
  - URL: arXiv

PDF Extraction Challenges & Solutions

- **PDF Hell and Practical RAG Applications** (LLMWhisperer Blog, 2025)
  - Addresses practical challenges of extracting text from PDFs for LLM/NLP applications
  - Discusses layout preservation challenges and character representation issues
  - URL: Unstract Blog (December 2025)
- **From PDFs to AI-Ready Structured Data: A Deep Dive** (Explosion AI)
  - Workflows for handling PDFs in NLP context
  - Covers layout analysis, OCR, table recognition, and structured output
  - Discusses transfer learning and fine-tuning approaches
  - Mentions Docling (IBM Research tool) for comprehensive PDF processing

    - URL: Explosion AI Blog
- **ChunkNorris: A High-Performance and Low-Energy Approach to PDF Parsing and Chunking**
    - Recent advances in PDF parsing techniques combining traditional and ML approaches
    - State-of-the-art methods (PDFMiner, PyPDF2 libraries)
    - URL: arXiv (2602.00010)
- **PDF Parsing With LLMs: Enterprise Guide** (LlamaIndex Blog)
    - LlamaParse for state-of-the-art vision-language processing
    - Addresses limitations of OCR and rule-based parsing
    - Enterprise-grade infrastructure for PDF transformation
    - URL: LlamaIndex.ai
- **Tabular PDF Information Extraction with Local LLMs and Layout-Aware Parsing** (2025)
    - Reliability evaluation of LLM-based extraction from tabular PDFs
    - URL: arXiv (2604.00003)

Information Extraction Fundamentals

- **Information Extraction with LLM** (Medium, 2023)
    - Uses advanced LLMs (Llama-v2) for extracting structured data from unstructured text
    - Discusses PDF complexities, OCR issues, and layout variations
    - Transformer-based architectures for capturing dependencies
    - URL: Medium (August 2023)

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3. MULTIMODAL LARGE LANGUAGE MODELS (VISION-LANGUAGE MODELS)

Capabilities & Comparisons

- **Multimodal AI Face-Off: Claude, GPT-4V, and Gemini in 2026**
    - Practical comparison of Claude 4.5, GPT-4V, and Gemini 3 for document analysis
    - Benchmarks across image understanding and vision-language tasks
    - URL: Claude 5 Hub (May 2026)
- **Claude 3.5 Sonnet Document Analysis Capabilities** (2026)
    - Excellent for reading PDFs, understanding form layouts, extracting table data
    - 200K context window for processing multiple images/documents

    - URL: LLM Trust (March 2026)
- **The Multimodal AI Guide: Vision, Voice, Text, and Beyond** (KDnuggets)
    - Comparison of GPT-4V, Claude, Gemini with open-source alternatives (LLaVA, Qwen-VL)
    - Discussion of OCR-free document understanding
    - URL: KDnuggets
- **Vision-Language Model Overview** (Grokipedia, 2026)
    - Advantages of multimodal VLMs over traditional OCR
    - Superior contextual understanding, layout awareness
    - URL: Grokipedia (March 2026)

Research Papers on Vision-Language Models

- **Document Haystack: A Long Context Multimodal Image/Document Understanding Vision LLM Benchmark**
    - Benchmarking multimodal LLMs (GPT-4, Gemini 1.5, LLama 3.2, Claude 3)
    - Importance of document understanding in legal, medical, financial sectors
    - URL: arXiv (2507.15882)
- **The Dawn of LMMs: Preliminary Explorations with GPT-4V(ision)** (Microsoft Corporation)
    - Comprehensive analysis of large multimodal models
    - Explores tasks, capabilities, prompting strategies
    - URL: arXiv (2309.17421)
- **GPT-4 Vision: Multi-Modal Evolution of ChatGPT and Potential Role in Radiology** (2024)
    - GPT-4V capabilities for text generation from images without specialized training
    - Transformation from LLM to Large Multimodal Model
    - URL: Cureus/NCBI
- **Multimodal Large Language Models: Transforming Computer Vision** (Edge AI Alliance, 2025)
    - Overview of MLLMs from major tech companies
    - Discusses GPT-4o, Claude 3.5 Sonnet, LLaVA, and open-source alternatives
    - URL: Edge AI and Vision Alliance (January 2025)
- **Multi-Agent Interactive Question Generation Framework for Long Document Understanding**
    - Multi-hop reasoning across extended documents
    - URL: arXiv (2507.20145)

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4. TOPIC MODELING & EXTRACTION

Latent Dirichlet Allocation (LDA) & Traditional Methods

- **Combining Language and Topic Models for Hierarchical Text Classification**
  - Overview of topic models (LDA, recent approaches like BERTopic)
  - LDA represents documents as distributions over topics
  - URL: arXiv (2507.16490)
- **Topic Modeling Algorithms: Learn About Mathematical Concepts** (Medium, 2023)
  - Comprehensive comparison of LDA, NMF, and BERTopic
  - Discusses probability distributions and decomposition approaches
  - URL: Medium (August 2023)

BERTopic & Modern Approaches

- **LDA Topic Modeling Tutorial with Python and BERTopic** (Holistic SEO, 2022)
  - BERTopic implementation using UMAP, HDBSCAN, Sentence Transformers
  - Practical guide for installation and usage
  - URL: Holistic SEO (October 2022)
- **Beyond Sentiment: Leveraging Topic Metrics for Political Stance Classification**
  - Comparison of LDA, NMF, and BERTopic coherence scores
  - BERTopic demonstrates superior coherence across datasets
  - Manual topic labeling based on extracted keywords
  - URL: arXiv (2310.15429)
- **AI-Powered Topic Modeling: Comparing LDA and BERTopic** (2024-2025)
  - BERTopic integration with ChatGPT-4-Turbo for automatic interpretation
  - Enhanced interpretability and semantic coherence vs. LDA
  - Compact, well-separated clusters from BERTopic
  - URL: NCBI/PMC and PubMed (February 2025)
- **Unravelling Technical Debt Topics Through Time, Programming Languages and Repository**
  - BERTopic's transformer-based approach for semantic meaning
  - Comparison with traditional methods (LDA, LSA)
  - URL: arXiv (2504.11714)
- **Moving Beyond LDA: A Comparison of Unsupervised Topic Modeling Techniques** (2024)
  - Advanced techniques: Sentence-BERT, UMAP, HDBSCAN, cTF-IDF
  - URL: arXiv (2412.14486)

- **Analyzing Folktales of Different Regions Using Topic Modeling and Clustering** (2022)
  - BERTopic's three-step process: embedding → dimensionality reduction → topic extraction
  - Overcomes LDA's limitations in capturing semantic relationships
  - URL: arXiv (2206.04221)

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5. LEARNING ANALYTICS DATASETS & BENCHMARKS

Open University Learning Analytics Dataset (OULAD)

- **Open University Learning Analytics Dataset** (Nature Scientific Data, 2017)
  - Comprehensive dataset containing 32,593 students across 22 module presentations
  - Includes student demographics and VLE interaction data from 2013-2014
  - Available on UCI Machine Learning Repository
  - URL: Nature Scientific Data & UCI Machine Learning Repository
- **Open Learning Analytics: A Systematic Review of OULAD Studies** (2022)
  - Review of benchmark studies using OULAD
  - Focus on assessing student performance in Virtual Learning Environments
  - URL: ACM Proceedings (2022)
- **ouladFormat R Package: Preparing OULAD for Analysis** (2025)
  - R package for loading and formatting OULAD
  - Simplifies preprocessing for educational research
  - URL: arXiv (2501.08366)
- **OULAD Datasets on Kaggle** (Updated regularly)
  - Multiple analysis examples: Random Forest models, Personalized Learning Paths, Engagement tracking
  - Community-contributed notebooks and case studies
  - URL: Kaggle Datasets

Predictive Modeling with OULAD

- **Predictive Modelling with OULAD: A Systematic Literature Review** (2024)
  - Reviews predictive approaches using OULAD
  - Covers algorithms: Logistic Regression, SVM, Random Forest, XGBoost
  - Emphasis on explainable AI for academic risk prediction
  - URL: ResearchGate (July 2024)

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6. LEARNING ANALYTICS DASHBOARDS & VISUALIZATIONS

Dashboard Design & Implementation

- **Everything You Need to Know About Learning Analytics Dashboard** (KITABOO, 2024)
  - Common visualizations: bar charts, line graphs, heatmaps, scatter plots
  - KPIs: engagement, progress, achievement, behavior
  - Personalization benefits for enhanced learner engagement
  - URL: KITABOO Blog (February 2024)
- **Learning Analytics Dashboard: A Tool for Providing Actionable Insights to Learners**
  - Design of dashboards for triggering positive behavioral changes
  - Prediction capabilities for learning outcomes based on trajectory
  - Challenges: determining effective visual elements without inducing anxiety
  - URL: Springer Nature (February 2022)
- **Development of the Learning Analytics Dashboard to Support Students' Learning Performance** (2014)
  - Tracks students' online behavior patterns in VLE
  - Uses data mining and visualization for comprehension
  - Usability testing with stimulus recall interviews
  - URL: JUCS Journal (October 2014)
- **LearningViz: A Dashboard for Visualizing, Analyzing and Closing Learning Performance Gaps** (2024)
  - User-centered design with interactive analytics
  - Supports instructors in detecting performance gaps and investigating factors
  - Emphasis on cognitive load reduction through visualization
  - URL: Springer Nature (December 2024)
- **The Current Landscape of Research and Practice on Visualizations and Dashboards for Learning Analytics** (2021)
  - Comprehensive overview of LAD research and practice
  - Design considerations: what data, for whom, why, and how to represent
  - Challenges: usability, evaluation, privacy, adoption
  - URL: ResearchGate (December 2021)
- **A Visual Dashboard to Track Learning Analytics for Educational Cloud Computing**
  - CloudTrail-Tracker for AWS usage analytics
  - Progress tracking and pending actions for student self-regulation

- Integration in multiple cloud subjects
- URL: NCBI/PMC

Related Research

- **AISSA: Implementation and Deployment of an AI-Based Student Slides Analysis Tool** (2025)
  - Learning analytics dashboards as key instruments for educational data
  - Evolution from log-based dashboards to richer multimodal approaches
  - Challenges: usability, evaluation, privacy, adoption
  - URL: arXiv (2605.04729)

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7. HUMAN-IN-THE-LOOP & INTERACTIVE MACHINE LEARNING

Foundational Concepts

- **Human-in-the-Loop Machine Learning: Active Learning and Annotation** (Manning/IEEE)
  - Practical guide to optimizing ML process with human feedback
  - Techniques: annotation, active learning, transfer learning
  - URL: IEEE Xplore (2024)
- **Human-in-the-Loop Machine Learning: A State of the Art** (Springer, 2022)
  - Categorizes human involvement: active learning, interactive ML, machine teaching
  - Discusses curriculum learning and explainable AI
  - URL: Artificial Intelligence Review (August 2022)
- **How to Build Human-in-the-Loop AI Pipelines Using Active Learning**
  - Maximizes human-machine partnership potential
  - Importance of human involvement throughout learning process
  - Wide applications: medical, retail, finance, agriculture, automotive, geospatial
  - URL: Humans in the Loop (February 2025)

Implementation Approaches

- **Human-in-the-Loop (HITL) Explained** (Encord Blog, 2026)
  - Combines human expertise with machine learning
  - Humans curate and annotate data before feeding to AI
  - HITL systems combine automated predictions with structured human oversight
  - URL: Encord (March 2026)

- **Human-in-the-Loop Systems for Adaptive Learning Using Generative AI** (NCSU, 2024)
  - Integrates student feedback into AI-generated solutions
  - Tagging techniques and prompt engineering for personalization
  - Retrieval-Augmented Generation (RAG) for educational material
  - URL: arXiv (2508.11062)
- **Interactive Training: Feedback-Driven Neural Network Optimization** (2024)
  - Rich body of work on human interventions during model training
  - Compares active learning, interactive ML, and machine teaching paradigms
  - Real-time, open-ended intervention during training
  - URL: arXiv (2510.02297)

Active Learning & Labeling

- **Efficient Human-in-the-Loop Active Learning** (2026)
  - Novel framework for data labeling with flexible query types
  - Addresses expensive and labor-intensive labeling, especially for specialized skills
  - Multi-objective approach to optimize expert time
  - URL: arXiv (2501.00277)
- **Human-in-the-Loop for Active Learning** (Humans in the Loop)
  - Surfaces instances close to decision boundaries
  - Identifies data drift in deployed AI models
  - Post-deployment review for bias and failure modes
  - URL: Humans in the Loop (February 2026)

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8. EXPLAINABILITY & INTERPRETABILITY IN AI/ML

SHAP (SHapley Additive exPlanations)

- **Explainable Artificial Intelligence Credit Risk Assessment** (2025)
  - SHAP and LIME frameworks for model interpretation
  - Model-agnostic approach working with any ML algorithm
  - Feature contribution analysis for predictions
  - URL: arXiv (2506.19383)
- **Understanding XAI: SHAP, LIME, And Other Key Techniques** (2025)
  - SHAP for both global and local interpretability
  - Rich visualizations: beeswarm plots, summary plots, dependence plots
  - Computational efficiency considerations

    - URL: AI Competence (June 2025)
- **A Perspective on Explainable AI: SHAP and LIME** (Wiley, 2024)
    - Comparison of SHAP and LIME frameworks
    - Interpretation framework highlighting strengths and weaknesses
    - URL: Advanced Intelligent Systems (June 2024)
- **Decoding AI Predictions: XAI with LIME, SHAP, and InterpretML** (Medium, 2024)
    - Practical guide to explainability techniques
    - Applications in regulatory compliance (GDPR, EU AI Act)
    - URL: Medium (May 2024)

LIME (Local Interpretable Model-Agnostic Explanations)

- **Interpreting AI Models: LIME and SHAP in Alzheimer's Disease Detection** (NCBI, 2024)
    - Systematic review of LIME and SHAP applications
    - Both frameworks effective for clinical decision support systems
    - URL: NCBI/PMC (2024)
- **Enhancing Interpretability via CNN-BiLSTM and SHAP Analysis** (2024)
    - Post-hoc interpretation methods (LIME, SHAP)
    - Permutation-based feature importance analysis
    - URL: arXiv (2512.06730)

Comprehensive XAI Reviews

- **Evaluating Explainability in AI Models: SHAP, LIME, and Other Techniques** (2024)
    - Compares effectiveness, computational efficiency, interpretability, applicability
    - Explores integrated gradients, counterfactual explanations, permutation importance
    - URL: ResearchGate (September 2024)
- **Mastering Explainable AI: SHAP, LIME, Counterfactuals, and Interpretable NNs** (Medium, 2024)
    - Comprehensive guide to XAI tools for transparent ML
    - Importance for building trust and compliance
    - URL: Medium (December 2024)
- **Towards Trustworthy Keylogger Detection via XAI** (2025)
    - Global and local perspectives of model decisions
    - SHAP summary plots and dependence plots
    - URL: arXiv (2505.16103)
- **Spectral Zones-Based SHAP/LIME** (NCBI/PMC, 2024)

- Enhanced interpretability through grouped feature analysis
- Addresses limitations of individual feature perturbations
- URL: NCBI/PMC (2024)

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9. PERSONALIZED LEARNING RECOMMENDATIONS & PATHS

Learning Path Optimization

- **Personalized Learning Path Optimization Based on Enhanced Deep Neural Network** (2025)
  - Deep learning and reinforcement learning for intelligent learning support systems
  - Real-time feedback and dynamic adjustments
  - Enhanced learner motivation and engagement
  - URL: Springer Nature (August 2025)
- **Personalized Learning Path Recommendation Based on Knowledge Graphs** (2026)
  - Knowledge graphs model concepts, courses, resources, and relationships
  - Evolution from content retrieval to multi-level personalization
  - Approaches: knowledge tracing, cognitive diagnosis, path optimization
  - URL: Electronics MDPI (January 2026)
- **Simulation of Personalized English Learning Path Using Knowledge Graphs and Deep RL** (2025)
  - Integrates domain knowledge graphs with deep reinforcement learning
  - Dynamic learner knowledge state perception
  - Recommendation strategy optimization
  - URL: Scientific Reports (October 2025)

Recommendation Systems & Strategies

- **Educational Personalized Learning Path Planning with Large Language Models** (2024)
  - LLM-based learning path planning for adaptive education
  - Recent advancements leveraging knowledge graphs and AI
  - Focus on interpretability of recommendations
  - URL: arXiv (July 2024)
- **Personalized Learning Path Planning Using Deep Generative Models and Quantum ML** (2025)
  - Transformer model for extracting features from multimodal data
  - Adversarial training for robustness

    - URL: Springer Nature (March 2025)
- **A Comprehensive Exploration of Personalized Learning in Smart Education** (2024)
    - Student modeling to personalized recommendations
    - Learning paths, course recommendations, exercise recommendations
    - Knowledge graph-based approaches
    - URL: arXiv (2402.01666)
- **Personalized Learning Path Recommendation Strategy Based on Subject Knowledge Graph** (2024)
    - Knowledge graphs as backbone for subject domain representation
    - Analyzing current knowledge state and identifying gaps
    - Dynamic adjustment and feedback loops
    - URL: ACM Proceedings (2024)
- **IB-GRPO: Multi-Objective Personalized Learning Path Recommendation** (2026)
    - LLM-based alignment with educational objectives
    - Optimization of multiple objectives simultaneously
    - URL: arXiv (2601.14686)

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10. USER EXPERIENCE & INTERFACE DESIGN FOR EDUCATION

Educational UI/UX Principles

- **UI/UX Design for Education: Enhancing Learning Experiences** (Medium, 2024)
    - User-centered approach for diverse learners
    - Empathy mapping and user personas for diverse needs
    - URL: Medium (October 2024)
- **Design for Education: Optimizing edTech Products** (Backpack Interactive, 2025)
    - Essential user flows: onboarding, rostering, content sequencing, dashboards
    - Actionable data dashboards for concrete next steps
    - Student agency and learner motivation
    - URL: Backpack Interactive (January 2025)
- **UX Design in Education for Student Engagement** (Lollypop Design, 2025)
    - Core principles: simplicity, clarity, engagement, usability
    - Balancing cognitive load with effective learning
    - URL: Lollypop Design (October 2025)

Dashboard & Interface Design

- **User-Centric Education UI/UX Design Services** (DreamX)

    - Specialized design for LMS, apps, student portals, virtual classrooms
    - Prioritizes information hierarchy and intuitive navigation
    - Mobile-responsive design for educational platforms
    - URL: DreamX Web
- **Design for Education: Complete UX Guide** (GitNexa, 2026)
    - UX sits at intersection of pedagogy, psychology, software design
    - UI (visual styling) vs. UX (how system works over time)
    - URL: GitNexa (April 2026)
- **Artificial Intelligence in Elementary STEM: Teacher Dashboard Design** (2024)
    - Aggregated indicators for teachers (engagement trends, attention heatmaps, participation)
    - Avoiding overwhelming educators or creating surveillance atmosphere
    - Design priorities based on pedagogical necessity
    - URL: arXiv (2511.00105)

KEY RESEARCH TAKEAWAYS FOR PROJECT

1. Document Parsing

- **Action**: Use multimodal LLMs (Claude 3.5 Sonnet, GPT-4V) for syllabus and PDF extraction
- **Why**: Superior OCR-free document understanding with contextual awareness
- **Tools**: pdfminer.six, PyPDF2 for baseline; LLM APIs for advanced parsing

2. Topic Extraction & Mapping

- **Action**: Implement BERTopic for question-to-topic mapping with AI verification
- **Why**: Better semantic coherence than LDA; integration with LLMs for automatic interpretation
- **Consider**: Human verification loop essential given ambiguous multi-topic questions

3. Performance Analytics

- **Action**: Use OULAD dataset patterns for validation; implement SHAP/LIME for explainability
- **Why**: Rich benchmark for educational ML; transparent scoring crucial for student trust
- **Model choice**: Rule-based scoring (primary) + Random Forest/Linear Regression (experimental)

- **Action**: Design around actionable insights, not data overwhelming
- **Why**: Learning analytics literature emphasizes clear, digestible visualizations
- **Key principle**: Show topic-wise accuracy trends, consistency scores, personalized recommendations

5. Human-in-the-Loop Verification

- **Action**: Central to project architecture (student verifies AI topic mappings)
- **Why**: Rich body of HITL research shows effectiveness in handling AI mistakes
- **Implementation**: Simple approve/edit interface reduces friction

6. Recommendation Engine

- **Action**: Personalize study recommendations using knowledge gaps + trends
- **Why**: Growing research on knowledge graph-based paths and LLM-generated recommendations
- **Approach**: Identify weak topics, rank by impact, generate actionable guidance

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ADDITIONAL RESOURCES

Datasets for Validation

- OULAD (Open University Learning Analytics Dataset): https://analyse.kmi.open.ac.uk/open_dataset
- Kaggle OULAD Variants: https://www.kaggle.com/datasets/

Relevant Libraries & Tools

- **Document Processing**: pdfminer.six, PyPDF2, Docling (IBM Research)
- **Topic Modeling**: BERTopic, scikit-learn LDA
- **ML Models**: scikit-learn (Random Forest, Linear Regression), XGBoost
- **Explainability**: SHAP, LIME
- **Visualization**: Recharts, D3.js, Plotly

Standards & Best Practices

- GDPR compliance for educational data
- Pedagogical principles for dashboard design
- Active learning and human-centered AI approaches

DOCUMENT METADATA

- **Compilation Date:** July 2026
- **Total Sources:** 100+ academic papers, blogs, and industry resources
- **Coverage:** EDM, ML prediction, document parsing, dashboards, HCI, XAI
- **Purpose:** Support literature review and project methodology

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HOW TO USE THIS DOCUMENT

1. **By Phase:** Group papers by project phases (PDF pipeline, analytics, ML, dashboard)
2. **By Topic:** Use sections to research specific areas (e.g., topic modeling for Phase 2)
3. **By Paper Type:** Mix academic papers (validation) with practical guides (implementation)
4. **For Your Report:** Cite papers relevant to your methodology, challenges, and design decisions
5. **For Team Discussion:** Reference papers to justify architectural decisions (e.g., why HITL matters)
